Jung-Sub Lim

Atmospheric Physicist · Cloud Microphysics & Climate

CIRES, University of Colorado Boulder / NOAA Chemical Sciences Laboratory

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Research Profile

Atmospheric physicist studying how droplets and ice crystals shape cloud evolution, precipitation, and climate. I combine observations, Lagrangian large-eddy simulation, stochastic dynamics, and scientific software across warm and mixed-phase cloud systems.

Appointments

2025-Present	**Postdoctoral Associate**, CIRES / NOAA CSL, University of Colorado Boulder
2021-2024	**Research Associate**, Meteorological Institute Munich / LMU
2019-2021	**Research Associate**, Geophysical Fluid Dynamics Lab., Dept. of Atmospheric Sciences, Yonsei University
2018	**Research Internship**, Computational Science and Engineering, Yonsei University
2015-2017	**Social Service**, Republic of Korea Army

Education

Ludwig-Maximilians-Universität München (LMU) — *Munich, Germany*

Dr. rer. nat. (Ph.D. Meteorology / Physics) — *Sep. 2021 – Dec. 2024*

- Dissertation title: Entrainment, Mixing, and the Evolution of the Cloud Droplet Size Distribution
- Honors: *magna cum laude*
- Committee: Prof. Dr. Fabian Hoffmann, Prof. Dr. Bernhard Mayer, Prof. Dr. George Craig, Prof. Dr. Dieter Braun

Yonsei University — *Seoul, Republic of Korea*

M.Sc. Atmospheric Sciences — *Sep. 2019 – Aug. 2021*

- Dissertation title: The Effects of Giant Aerosol and Turbulence on the Stochastic Coalescence in Clouds
- Committee: Prof. Dr. Yign Noh, Prof. Dr. Hyunho Lee, Prof. Dr. Sungsoo Yum

Yonsei University — *Seoul, Republic of Korea*

B.Sc. Atmospheric Sciences — *Sep. 2017 – Aug. 2019*

Pusan National University — *Busan, Republic of Korea*

Undergraduate Studies in Atmospheric Sciences — *Mar. 2012 – Dec. 2014*

- Transferred to Yonsei University

Publications

Published

[5] Zhang, J., Painemal, D., Dror, T., **Lim, J. S.**, Sorooshian, A., & Feingold, G. (2026). Inferring processes governing cloud transition during mid-latitude marine cold-air outbreaks from satellite. ***Atmospheric Chemistry and Physics***, 26, 6015-6034. doi:10.5194/acp-26-6015-2026

[4] **Lim, J. S.**, & Hoffmann, F. (2026). Aging of droplet size distribution in stratocumulus clouds: regimes of droplet size distribution evolution. ***Atmospheric Chemistry and Physics***, 26, 5427–5446. doi:10.5194/acp-26-5427-2026

[3] **Lim, J. S.**, Noh, Y., Lee, H., Hoffmann, F. (2025). The Critical Number and Size of Precipitation Embryos to Accelerate Warm Rain Initiation. ***Atmospheric Chemistry and Physics***, 25(10), 5313–5329. doi:10.5194/acp-25-5313-2025

[2] **Lim, J. S.**, & Hoffmann, F. (2024). Life cycle evolution of mixing in shallow cumulus clouds. ***Journal of Geophysical Research: Atmospheres***, 129(10), e2023JD040393. doi:10.1029/2023JD040393

[1] **Lim, J. S.**, & Hoffmann, F. (2023). Between Broadening and Narrowing: How Mixing Affects the Width of the Droplet Size Distribution. ***Journal of Geophysical Research: Atmospheres***, 128(8), e2022JD037900. doi:10.1029/2022JD037900

Presentations

** presenting author*

Invited Talks

From Particle to Planet: What Cloud Droplets Tell Us About Arctic Climate. Korea Institute of Science and Technology, Seoul, Republic of Korea., May, 2026

Stochastic Cloud Fluctuations and Arctic Winter Radiative Bistability: Bridging Cloud Microphysics and Climate. Kyungpook National University, Daegu, Republic of Korea., May, 2026

Evolution of Individual Cloud Droplets: A Lagrangian View of Turbulent Entrainment, Mixing and Droplet Growth Pathways. Ewha Womans University Colloquium, Seoul, Republic of Korea., Oct., 2025

Scale-Crossing Lagrangian Cloud Modeling: From the Kilometer Scale to the Millimeter Scale. Ewha Womans University Colloquium, Seoul, Republic of Korea., Oct., 2022

Media and Press

SBS 8 O'Clock News (29.08.2023) Featured interview on research regarding cloud–climate interactions in *SBS 8 O'Clock News* (Aug. 29, 2023). `https://www.youtube.com/watch?v=bbkcWEUPxS0`

Contributed Presentations

Lim, J. S., and Feingold, G. 2026., Cloud-Driven Stochastic Variability Sustains Radiative Bimodality in the Arctic Winter, Poster presentation: GML 53rd Global Monitoring Annual Conference, Boulder, Colorado

Lim, J. S., and Hoffmann, F. 2026. Regimes of Droplet Size Distribution Evolution in Stratocumulus: From Adiabatic Growth to Entrainment–Descent. Oral presentation: Third Symposium on Cloud Physics at the 106th AMS Annual Meeting, Houston, Texas

Kim S., La, I, Hoffmann, F., **Lim, J. S.**, and Yum, S., 2025., Modeling Inhomogeneous and Homogeneous Mixing Traits in Marine Stratocumulus Clouds: Effects of CTEI and Radiative Cooling. Presentation: Third Symposium on Cloud Physics at the 106th AMS Annual Meeting, Houston, Texas

Rug, L., Ascher, B., Kainz, J., **Lim, J. S.**, and Hoffmann, F. 2025., Marine Cloud Brightening: Constraining Small-Scale Uncertainty. ICNAA 2025, Vienna, Austria

Kazil, J., Vogel, R., Alinaghi, P., Bariteau, L., Bayley, C., Blossey, P., ..., **Lim, J. S.**, ... & Zuidema, P. Cold Pool Analysis from The Cold Pool Model Intercomparison Project (CP-MIP). 105th Annual AMS Meeting, New Orleans, 2025.

Lim, J. S., and Hoffmann, F. 2025., Regimes of Droplet Size Distribution Evolution in Stratocumulus: From Adiabatic Growth to Entrainment and Mixing. Poster Presentation: EGU General Assembly 2025, Vienna, Austria

Lim, J. S., and Hoffmann, F. 2025., Aging of DSD in Stratocumulus: The Role of Adiabatic Growth and Entrainment-Mixing. Oral Presentation: Workshop on Cloud Across Scales, Zugspitze, Bavaria, Germany, April 2025

Lim, J. S., and Hoffmann, F., 2024. Life Cycle Evolution of Inhomogeneous Mixing in Shallow Cumulus Clouds. Oral Presentation: ICCP 2024, Jeju, Republic of Korea

Lim, J. S., and Hoffmann, F., 2024. Case D: Precipitating cumulus congestus cloud with SAM-LCM. Oral Presentation: ICMW 2024, Seoul, Republic of Korea

Lim, J. S., and Hoffmann, F., 2024. Environmental and Lifecycle Effects on Entrainment and Mixing in Maritime Shallow Cumulus Cloud. Oral Presentation: ACPC 2024, London, United Kingdom

Lim, J. S., and Hoffmann, F., 2023. The Evolution from Homogeneous to Inhomogeneous Mixing During Cloud Lifecycle. Poster Presentation: AGU 2023 Fall meeting, San Francisco, USA

Lim, J. S., 2023. Scale-Crossing Lagrangian Cloud Modeling to Understand the Role of Clouds in Climate. Oral Presentation: The Korean Scientists and Engineers Association in Germany 2023 Fall meeting, Essen, Germany

Lim, J. S. Effect of Anthropogenic Aerosols on Cloud Lifecycle. Oral Presentation: Europe-Korea Conference on Science and Technology 2023, Munich, Germany

Lim, J. S., and Hoffmann, F., 2023. Between Broadening and Narrowing: How Mixing Affects the Width of the Droplet Size Distribution. Poster Presentation: 2023 CFMIP-GASS meeting, Paris, France

Lim, J. S., and Hoffmann, F., 2022. Effects of Entrainment and Mixing Scenarios on the Droplet Size Distribution: Spatiotemporal Variability and Cloud-Lifecycle Dependence. Poster Presentation: AGU 2022 Fall Meeting, Chicago, USA

Lim, J. S., Kainz, J., and Hoffmann, F., 2022. The Present and Future of Lagrangian Cloud Modeling: From the Centimeter to Kilometer Scale. Oral Presentation: EMS 2022 Annual meeting, Bonn, Germany

Lim, J. S., Hoffmann, F., 2022. Entrainment and Mixing in Cumulus Clouds: A New Mixing Scenario That Narrows the Width of the Droplet Size Distribution. Oral Presentation: AMS, 16th Conference on Cloud Physics, Madison, USA

Lim, J. S. The Effects of Giant Aerosols in Clouds and their Possible Role in Geoengineering. Poster Presentation: Europe-Korea Conference on Science and Technology 2022, Marseille, France

Lim, J. S., Noh, Y., Hoffmann, F., Lee, H., 2021. Idealized Lagrangian Cloud Model Approach to Investigate Early Collisional Growth of Cloud Droplet: Effect of Giant Aerosol and Turbulence. Poster Presentation: AGU 2021 Fall meeting, Online

Lim, J. S., Lee, H., Noh, Y., 2020. Impact of GCCN and TICE on the formation of precipitation and lucky droplets using LCM ensemble model. Oral Presentation: Korean Meteorological Society Fall session, Online

Kim, S., **Lim, J. S.**, Choi, J., 2018 Optimal Ventilation Efficiency Analysis in Various Circular Building Structures Using Immersed Boundary Methods, Oral Presentation: Korean Society of Mechanical Engineering Spring Annual Conference, Jeongseon, Republic of Korea

Research Program & Scientific Software

Particle histories and cloud evolution. Study how growth, mixing, collision, and transport reshape droplets and ice crystals over their lifetimes and influence cloud structure and precipitation.

Cloud transitions and climate. Investigate how microphysics and variability govern whether clouds rain, change phase, persist, or dissipate under changing environmental conditions.

Lagrangian modeling across scales. Develop particle-based models and model–observation comparisons that connect individual hydrometeor histories with cloud evolution and climate-relevant processes.

DropLab. Creator and maintainer of an open-source Lagrangian super-droplet laboratory spanning warm-rain parcels, idealized 2-D warm and mixed-phase clouds, deep convection, and climate-intervention demonstrations. `https://github.com/jslim93/DropLab`

GPU-accelerated particle-based microphysics in LES. Developing GPU-accelerated particle-based microphysics for the Lagrangian Cloud Model (LCM), coupled to the System for Atmospheric Modeling (SAM), using OpenACC on NVIDIA H100 GPUs together with a physics-validation workflow. Current end-to-end benchmarks span roughly 7–30× across cases and scales; performance characterization and statistical validation are ongoing.

Teaching Experience

Lectures

2026	**Cloud Modelling from a Lagrangian Perspective**, Guest Lecturer	*Kyungpook National University*
2024	**Introduction to Scientific Writing**, Invited Lecturer	*VeKNI*
2022	**Cloud–Climate Interactions**, Invited Speaker	*VeKNI*

Teaching Assistant

Spring 2023	**Instrumenten-Praktikum (Meteorological Instruments Practicum)**, TA	*LMU*
Spring 2022	**Instrumenten-Praktikum (Meteorological Instruments Practicum)**, TA	*LMU*
Fall 2020	**Fluid dynamics**, TA	*Yonsei University*
Spring 2020	**Climate and civilization**, TA	*Yonsei University*

Mentored Students

since 2025	**Sanggyeom Kim**, Ph.D. Student	*Yonsei University, Republic of Korea*
2023	**Julian Sebastian Humer-Hager**, student research assistant	*LMU, Germany*

Research Funding

Pending	**Causal Understanding of the Transition between Radiatively Clear and Opaque Arctic States**, DOE BER/ASR; Co-Investigator	*$941k*

Awards & Fellowships

2019, 2020, 2021	**BK21 (Brain Korea 21) Graduate Fellowship**, NRF Korea
2018	**Bronze Prize in the Undergraduate Fluid Dynamics Competition**, Korean Society of Mechanical Engineering
2014	**Undergraduate scholarship**, The East
2013, 2014	**Undergraduate scholarship**, Rotary International
2013	**Undergraduate scholarship**, Shin-heung Youth Association

Outreach & Professional Development

Professional Service

Lead Convener and Chair, "Mixed-Phase Clouds: Bridging Observations, Laboratory Studies, Modeling, and Theory," Fourth Symposium on Cloud Physics, 107th AMS Annual Meeting (accepted; 2027).

Lead Convener and Chair, two mixed-phase-cloud sessions at the Third Symposium on Cloud Physics, 106th AMS Annual Meeting (2026).

Peer Reviewer

15 completed reviews for journals including Quarterly Journal of the Royal Meteorological Society, Journal of Geophysical Research: Atmospheres, Journal of Atmospheric Sciences, npj Climate and Atmospheric Science, Atmospheric Research, Atmospheric Chemistry and Physics, Geophysical Research Letters, Journal of Advances in Modeling Earth Systems, GIScience & Remote Sensing, and Weather.

Membership

Korean Meteorological Society (since 2020), American Geophysical Union (since 2021), American Meteorological Society (since 2022), The Korean Scientists and Engineers Association in the FRG (VeKNI) (2022-2024)

Service and Outreach

2025-2026	**CSL Science Social All-hands**, Organizing committee member	*NOAA CSL*
2020-2021	**Youth meteorology career camp for high school and middle school students**, Lecturer	*Korea Productivity Center*
2019	**Weather Debate Contest: Finalist team (top 8 teams)**, Team leader	*Korea Meteorological Administration*
2018-2019	**Undergraduate Student Society of Atmospheric Sciences**, Co-founder/University representative	*Yonsei-SNU-Ewha joint group*
2018	**Big Data Study Club, BITAMIN**, Member/Python teaching member	*Seoul, Korea*